

Historical Persistence & Geospatial Economics

Applied Economics Research Course

Bas Machielsen
Utrecht University
2025-04-23

Introduction

Institutions

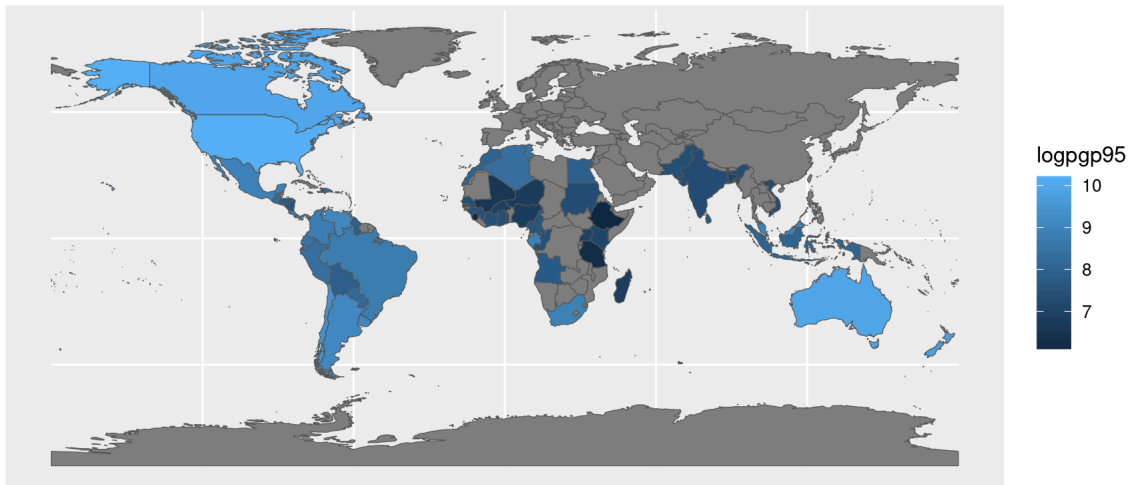
- Economic or other outcomes are not only influenced directly by relative prices, costs and benefits.
- They are also influenced by more latent, long-term factors, often called institutions.
 - Example: Acemoglu, Robinson and Johnson (2001):

(..) estimate the effect of institutions on economic performance. Europeans adopted very different colonization policies in different colonies, with different associated institutions. In places where Europeans faced high mortality rates, they could not settle and were more likely to set up extractive institutions. (..) Exploiting differences in European mortality rates as an instrument for current institutions, we estimate large effects of institutions on income per capita.

Institutions

- Here's a world map with GDP/capita in 1995 for a selected number of countries:

```
library(sf); library(rnaturalearthdata); library(ivdoctr)  
world_map ← rnaturalearthdata::countries50 ▷ st_as_sf()  
wm ← world_map ▷ left_join(ivdoctr::colonial,  
                           by = c("iso_a3" = "shortnam"))  
wm ▷ ggplot(aes(fill=logpgp95)) + geom_sf()
```



Institutions

- In this paper, Acemoglu, Johnson and Robinson (2001) take institutions to be *expropriation risk*: Measures risk of government appropriation of foreign private investment on a scale from 0 (least risk) to 10 (most risk), averaged over all years from 1985-1995.
- If one wants to find the relationship between institutions and economic well-being, one runs into the problem of *endogeneity*:
 - Institutions causally affects economic growth
 - But economic growth also affects institutions!
 - Formally:
 - $\text{GDP/Cap}_i = \alpha_0 + \alpha_1 \text{Institutions}_i + \epsilon_i^1$
 - $\text{Institutions}_i = \beta_0 + \beta_1 \text{GDP/Cap}_i + \epsilon_i^2$
- If this is the *true* data-generating process, an OLS regression will not yield the correct coefficient you are interested in (α_1).

Instrumental Variables

- People have devised various strategies to solve this problem. The most often-used strategy is to use an *instrumental variable*
 - An instrumental variable is a variable that exogenously causes a change in $\mathbf{X}$ while not directly affecting $\mathbf{Y}$.
 - In the AJR (2001) case:
$$\text{Institutions}_i = \beta_0 + \beta_1 \text{GDP/Cap}_i + \underbrace{\beta_2 \text{SettlerMortality}_i}_{\text{Instrumental Variable}} + \epsilon_i^2$$
 - Now, GDP per capita and Institutions are *endogenously* determined as a result of *exogenous* settler mortality

Instrumental Variables

- You can rearrange these two equations to find that:

$$\circ \text{Institutions}_i = \dots + \underbrace{\left(\frac{\beta_1 \alpha_1 \beta_2}{1 - \alpha_1 \beta_1} + \beta_2 \right)}_{= \frac{\beta_2}{1 - \alpha_1 \beta_1}} \cdot \text{SettlerMortality}_i + \dots$$

$$\circ \text{GDP/Cap}_i = \dots + \left(\frac{\alpha_1 \beta_2}{1 - \alpha_1 \beta_1} \right) \cdot \text{SettlerMortality}_i + \dots$$

- Now you can find (*identify*) α_1 by dividing $\text{Cov}(\text{GDP/Cap}, \text{SettlerMort})$ by $\text{Cov}(\text{SettlerMort}, \text{Institutions})$
- The key (and untestable) assumption here is that settler mortality does not directly influence GDP per capita in 1995.
- Loosely speaking, settler mortality represents an exogenous shock to institutions, allowing us to identify the influence of a change in institutions on GDP per capita

Historical Persistence

Back to historical persistence

- In general, we are dealing with outcomes that are determined endogenously: outcomes now and outcomes in the past are determined by other, latent factors
- To identify the influence of these latent factors, economists often use large-scale, influential events that shock these institutions. They compare outcomes in places that were initially similar, but some of them have been coincidentally exposed to certain events, whereas others have not.
- Usually, this research is based on spatial regression discontinuity designs, comparing places at one side of the border with places at the other side of the border (e.g. Dell, 2010, Lowes and Montero, 2021).

Historical Persistence

- The set-up in general is:
 - Outcome Y_i has “deep roots” caused by some D_i long ago
 - This D_i usually did not come to be randomly, but at the margin near some threshold, it did
 - Example (Dell 2011): Spanish conquest of Latin America was bounded by mountainous areas
- Comparing villages just within vs. just outside of the old imperial borders
- Other seminal papers:
 - Acemoglu et al. (2011), Occupied areas by Napoleon in Germany do better
 - Dell & Olken (2020): Positive development effects of extractive colonialism in Indonesia
 - Voigtlander & Voth (2012): Deep roots of persecution in Nazi Germany

Spatial Regression Discontinuity

- Consider the example of **Roman roads**. Let's suppose that Roman roads, by influencing trade networks, stimulate development:

- $$Y_i = \alpha_0 + \alpha_1 \text{RomanRoad}_i + \alpha_2 \text{Environment}_i + \epsilon_i^1$$

- But Roman Roads have not been built randomly by the Romans, but might have built in areas with more suitable environments for development:

- $$\mathbb{P}[\text{RomanRoad}_i] = \beta_0 + \beta_1 \text{Environment}_i + \epsilon_i^2$$

- Then, comparing areas with roads to areas without roads doesn't work due to differences in the environment:

- $$\mathbb{E}[Y | RR = 1] = \alpha_0 + \alpha_1 + \alpha_2 \mathbb{E}[\text{Environment} | RR = 1]$$

- $$\mathbb{E}[Y | RR = 0] = \alpha_0 + \alpha_2 \mathbb{E}[\text{Environment} | RR = 0]$$

Identification Strategy

- If we can find two places with the same expected value of getting a road, but only *one* of them gets a road..
- According to the equations on the previous slide, this means two places with the same environment (take the expected value of Y_i conditional on Environment to see that)
- Then, we can estimate α_1 by:
 - $\mathbb{E}[Y|\text{RomanRoad} \ \& \ P[\text{RomanRoad}]] - \mathbb{E}[Y|\text{NoRomanRoad} \ \& \ P[\text{RomanRoad}]]$
- This is what we attempt to do in a spatial regression discontinuity design: compare two arguably identical places with the same probability of treatment, where only one of them actually gets the treatment

The Data

Data

- I have two a couple of data sets on offer to you:
 - A dataset of French and German municipalities overlaid with Roman roads
(`france_germany.shp`)
 - This dataset contains the distance of each municipality to a Roman road, as well as an indicator, whether a road stretches through a municipality or not
 - A dataset of Dutch municipalities overlaid with the border of the former Roman empire (`netherlands_rome.geojson`)
 - Again including a variable for distance to the border, and an indicator of whether a present-day municipality was inside or outside the former Roman empire
- You are also free to use or compile a custom dataset using different settings and countries.

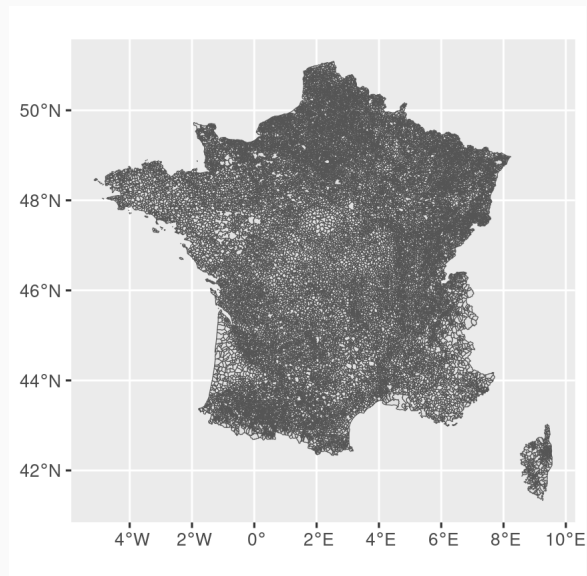
Demonstration

Demonstration

```
library(cawd); library(giscoR)
roads ← cawd::darmc.roman.roads.major.sp ▷ st_as_sf()
lau ← giscoR::gisco_get_lau(year = '2019', country = c("France"))

xmin ← -10; xmax ← 20; ymin ← 40; ymax ← 80
box ← st_bbox(c(xmin=xmin, xmax=xmax, ymax=ymax, ymin=ymin), crs = st_crs(4326))
square ← st_sf(st_as_sfc(box))
fr ← st_intersection(lau, square)

fr ▷ ggplot() + geom_sf()
```



What does this data.frame look like?

```
fr ► head(5)
```

```
## Simple feature collection with 5 features and 10 fields
```

```
## Geometry type: POLYGON
```

```
## Dimension:      XY
```

```
## Bounding box:   xmin: 4.248763 ymin: 49.54231 xmax: 5.31495 ymax: 50.12165
```

```
## Geodetic CRS:   WGS 84
```

##		id	GISCO_ID	CNTR_CODE	LAU_ID	LAU_NAME	POP_2019	POP_DENS_2019	AREA_KM2
## 1	FR_08026	FR_08026		FR	08026	Aubigny-les-Pothées	318	21.48498	14.8010
## 2	FR_08027	FR_08027		FR	08027	Auboncourt-Vauzelles	102	19.06222	5.3508
## 3	FR_08028	FR_08028		FR	08028	Aubrives	874	82.53602	10.5893
## 4	FR_08029	FR_08029		FR	08029	Auflance	85	13.92512	6.1040
## 5	FR_08030	FR_08030		FR	08030	Auge	61	13.42680	4.5433

```
##           _ogr_geometry_
```

```
## 1 POLYGON ((4.404236 49.75043 ...
```

```
## 2 POLYGON ((4.465646 49.56644 ...
```

```
## 3 POLYGON ((4.770839 50.09282 ...
```

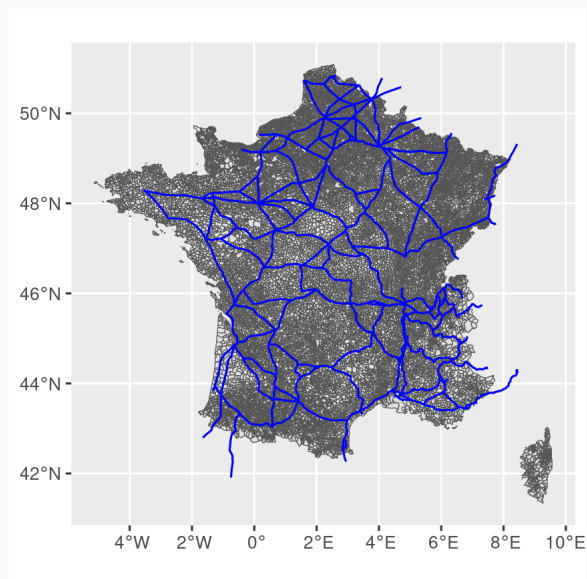
```
## 4 POLYGON ((5.289269 49.62845 ...
```

```
## 5 POLYGON ((4.260257 49.86763 ...
```

Merge this with the roads

```
numbers <- st_intersects(roads, fr)▷  
  as.data.frame() ▷ select(row.id) ▷ pull() ▷ unique()  
  
roads_in_fr <- roads ▷ filter(is.element(row_number(), numbers))
```

```
ggplot() +  
  geom_sf(data = fr) +  
  geom_sf(data = roads_in_fr, color = 'blue')
```



Compute the distance

```
minimum_distances <- fr ▷  
  st_centroid() ▷  
  st_distance(roads_in_fr) ▷  
  apply(1, min)
```

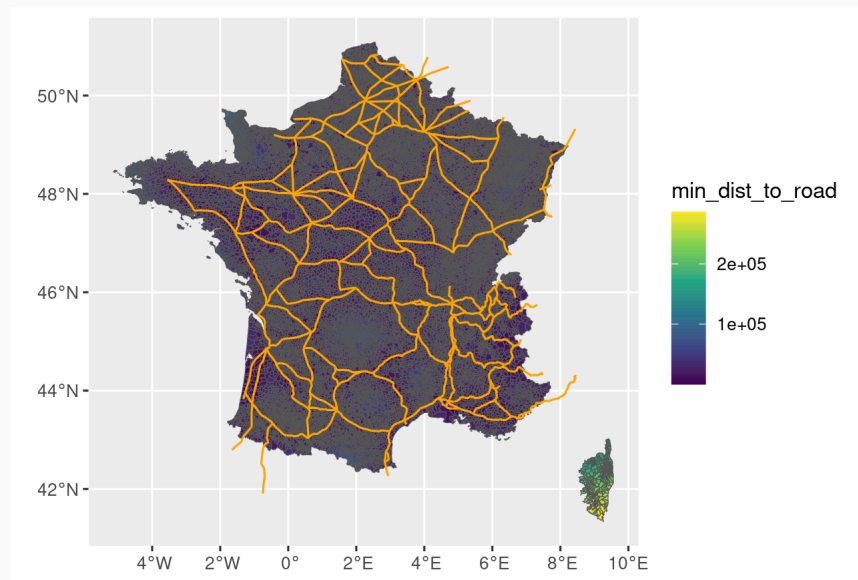
```
minimum_distances[1:10]
```

```
## [1] 12389.074  5728.599 34012.564  7624.885 24728.569 11728.691  2126.628  7900.327  8
```

```
fr <- fr ▷  
  mutate(min_dist_to_road = minimum_distances)
```

Plot the result

```
ggplot() +  
  geom_sf(data = fr, aes(fill = min_dist_to_road), size = 0.0001) +  
  scale_fill_viridis_c() +  
  geom_sf(data = roads_in_fr, color = 'orange')
```



A small analysis

```
library(fixest); library(modelsummary)
model1 <- fixest::feols(POP_DENS_2019 ~ min_dist_to_road, data = fr)
modelsummary(model1, gof_map = c("r.squared", "nobs"), stars=T)
```

	 (1)
(Intercept)	195.676***
	(4.821)
min_dist_to_road	-0.002***
	(0.000)
R2	0.004
Num.Obs.	35227
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	

Your task

- Find municipality-level or geographical outcomes that suit an *interesting research question*
 - Present-day or more historical/macro-level outcomes
- Make sure you control for environments!
 - Overlapping regions (provinces, supersets of municipalities)
 - But also possible: terrain, other initial conditions (coastal municipalities)
 - Make it plausible that you compare apples with apples
 - Some roads have also been built with a destination in mind: how to control for that?
- Explore mechanisms through which the effect can work

Geospatial Economics

Introduction

- In economics, many research questions can be answered using geospatial data. For example, questions related to economic development (Beyer et al., 2021; Besley et al., 2022) make frequent use of variation between different geographic units and compare them in aspects such as nightlight density and electricity consumption.
- Similarly, questions related to environmental economics can also be answered using similar data on the basis of geospatial variation (Castells-Quintana et al., 2021; Felbermayer et al., 2022).
- This theme will focus on similar research questions with the aim of introducing students to geospatial data wrangling and econometric methods.

Example

- This theme is a little bit broader than the Historical Persistence theme
- Many research questions are possible.
- Usually, your outcome variable is based on a *shapefile* containing an interesting outcome, e.g. nightlights, species distribution, pollution
- As an example, I have prepared a dataset and analysis of the following research question:
 - Does the air quality influence the performance of French schools?

Loading the Data

- Load the school performance dataset (source: data.gouv.fr)

```
school_performance ← read_delim(' ../ ../data/schools/fr-en-indicateurs-valeur-ajoutee-
```

- Filter so that the year is 2022:

```
school_performance ← school_performance ▷  
  filter(Session = 2022)
```

Geocoding the Schools

- We have to geocode the schools using a dataset containing the coordinates of these schools:

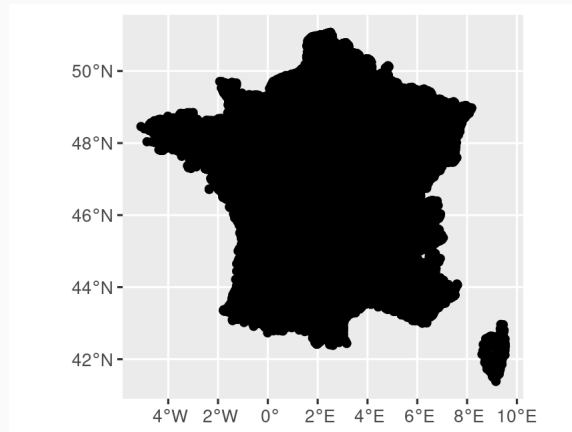
```
coordinates <- read_delim(' ../ ../data/schools/fr-en-adresse-et-geolocalisation-etabli:
  select(numero_uai, latitude, longitude) ▷
  filter(!is.na(latitude), !is.na(longitude)) ▷
  st_as_sf(coords=c('longitude', 'latitude'), crs=4326)
```

Filtering

- Filter only Metropolitan France (for the sake of visualization)

```
library(geodata)
france <- geodata::gadm("France", path=tempdir()) ▷
  st_as_sf()
coordinates <- coordinates ▷
  st_filter(france)

coordinates ▷
  ggplot() + geom_sf()
```



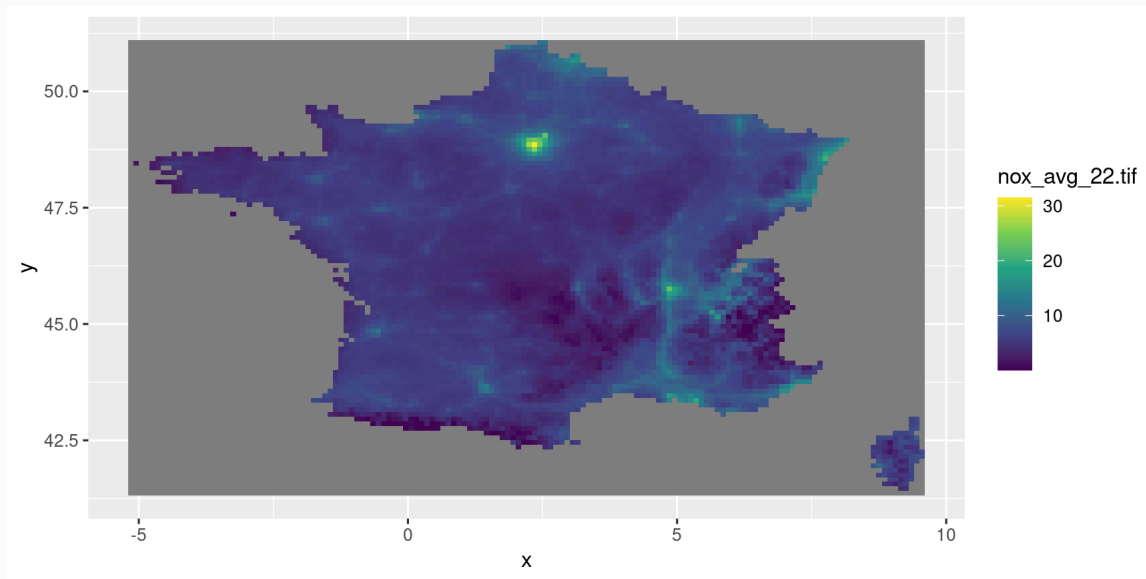
Importing Pollution Data

- This is annual NOx concentration data (Source: [European Environment Agency](#)):
 - NOx stands for Nitrogen Oxides—a group of harmful gases (mainly NO and NO₂) produced when fossil fuels (like gasoline, diesel, coal) burn at high temperatures. Common sources are cars & trucks, power plants and factories
- Read the file, downsample it (to keep it computationally manageable)
- Put it in the same CRS, keep only raster squares that are in France:

```
library(stars); library(dplyr)
sf_use_s2(F)
nox ← read_stars("../.. /data/schools/nox_avg_22.tif")
nox_lowres ← nox ▷ st_transform(st_crs(france))
nox_lowres ← st_warp(src=nox_lowres, cellsize = c(0.1, 0.1), crs=4326)
nox_lowres ← nox_lowres ▷ st_crop(st_union(france))
```

Plot Result

```
ggplot() + geom_stars(data=nox_lowres) + scale_fill_viridis_c()
```



Link Coordinates to Schools

```
geocoded_schools <- inner_join(coordinates, school_performance, by = c('numero_uai'='l  
geocoded_schools > head(5)
```

```
## Simple feature collection with 5 features and 29 fields
```

```
## Geometry type: POINT
```

```
## Dimension:      XY
```

```
## Bounding box:  xmin: -2.097861 ymin: 47.83106 xmax: -1.19397 ymax: 48.34177
```

```
## Geodetic CRS:  WGS 84
```

```
## # A tibble: 5 × 30
```

```
##   numero_uai          geometry num_ligne Session `Nom de l'établissement`  Commune  
##   <chr>              <POINT [°]>    <dbl>    <dbl> <chr>                        <chr>  
## 1 0350759K      (-1.68343 48.11588)    1289     2022 COLLEGE ECHANGE          RENNES  
## 2 0350762N      (-1.19397 48.34177)    1251     2022 COLLEGE THERESE PIERRE    FOUGERES  
## 3 0350868D      (-1.3157 47.83106)    1269     2022 COLLEGE PRIVE ST JOSEPH    MARTIGNE  
## 4 0350869E      (-1.993239 47.89385)    1327     2022 COLLEGE PRIVE STE MARIE    VAL D ANA  
## 5 0350878P      (-2.097861 47.99924)    1284     2022 COLLEGE PRIVE DE L'HERMINE PLELAN LE  
## # i 23 more variables: `Région académique` <chr>, `Code académie` <chr>, Académie <chr>,  
## #   Département <chr>, Secteur <chr>, `Nb candidats G` <dbl>, `Taux de réussite G` <dbl>,  
## #   `Nb candidats P` <dbl>, `Taux de réussite P` <dbl>, `Note à l'écrit G` <dbl>, `VA de  
## #   `Note à l'écrit P` <dbl>, `Taux d'accès 6eme 3eme` <dbl>, `Part présents 3eme ordina  
## #   `Part présents 3eme ordinaire G` <dbl>, `Part présents 3eme ordinaire P` <dbl>, `Par  
## #   `Nb mentions AB G` <dbl>, `Nb mentions B G` <dbl>, `Nb mentions TB G` <dbl>, `Nb men
```

Link Geocoded Schools to Pollution

```
pollution_values <- st_extract(nox_lowres, geocoded_schools, FUN=mean, na.rm=T)
geocoded_schools <- geocoded_schools >
  mutate(pollution_values=pollution_values$nox_avg_22.tif)
geocoded_schools > head(5)
```

```
## Simple feature collection with 5 features and 30 fields
```

```
## Geometry type: POINT
```

```
## Dimension: XY
```

```
## Bounding box: xmin: -2.097861 ymin: 47.83106 xmax: -1.19397 ymax: 48.34177
```

```
## Geodetic CRS: WGS 84
```

```
## # A tibble: 5 × 31
```

```
##   numero_uai          geometry num_ligne Session `Nom de l'établissement` Commune
##   <chr>          <POINT [°]>    <dbl>    <dbl> <chr>          <chr>
## 1 0350759K      (-1.68343 48.11588)    1289    2022 COLLEGE ECHANGE      RENNES
## 2 0350762N      (-1.19397 48.34177)    1251    2022 COLLEGE THERESE PIERRE  FOUGERES
## 3 0350868D      (-1.3157 47.83106)    1269    2022 COLLEGE PRIVE ST JOSEPH  MARTIGNE
## 4 0350869E      (-1.993239 47.89385)    1327    2022 COLLEGE PRIVE STE MARIE  VAL D ANA
## 5 0350878P      (-2.097861 47.99924)    1284    2022 COLLEGE PRIVE DE L'HERMINE PLELAN LE
## # i 24 more variables: `Région académique` <chr>, `Code académie` <chr>, Académie <chr>,
## #   Département <chr>, Secteur <chr>, `Nb candidats G` <dbl>, `Taux de réussite G` <dbl>,
## #   `Nb candidats P` <dbl>, `Taux de réussite P` <dbl>, `Note à l'écrit G` <dbl>, `VA de
## #   `Note à l'écrit P` <dbl>, `Taux d'accès 6eme 3eme` <dbl>, `Part présents 3eme ordina
## #   `Part présents 3eme ordinaire G` <dbl>, `Part présents 3eme ordinaire P` <dbl>,
## #   `Nb mentions AB G` <dbl>, `Nb mentions B G` <dbl>, `Nb mentions TB G` <dbl>, `Nb me
```


Small Analysis

- Clean up variable names, and run a regression

```
library(fixest); library(janitor)
geocoded_schools <- geocoded_schools %>%
  clean_names()
```

```
model <- feols(va_de_la_note_g ~ pollution_values + nb_mentions_tb_g, data=geocoded_schools)
modelsummary(model, gof_map = c("r.squared", "nobs"), stars=T)
```

	 (1)
(Intercept)	-0.001
	(0.020)
pollution_values	-0.012***
	(0.002)
nb_mentions_tb_g	0.004***
	(0.000)
R2	0.023
N	5000

Logistics

Course Schedule

- Upload research plan in Osiris 13 May 2025 at the latest
- Complete paper and upload in Osiris 26 June 2025 at the latest

Event	Date	Day
Lecture 1	23-04	Wednesday
Lecture 2	30-04	Wednesday
Lecture 3	07-05	Wednesday
Lecture 4	12-05	Monday
Lecture 5	28-05	Wednesday
Lecture 6	02-06	Monday
Lecture 7	11-06	Wednesday
Lecture 8	16-06	Monday

Lecture Logistics

- Lecture 1: Plenary Lecture, Introduction
- Lecture 2: Discussion of RQs (Send them on Teams!)
- Lecture 3 + 4: Plenary Lectures, Geospatial Data
 - Depending on time: room to deliberate
- Lectures 5-7: Feedback sessions, room to work and discuss
- Lecture 8: Presentations, feedback from supervisor and co-reader

Expected output

- At the end of week 3 you will upload your research plan in Osiris, including problem statement, (preliminary) literature review and research approach. This plan and your performance in the research group so far will be the input for your supervisors' go/no go decision (crucial with respect to your compliance with the effort requirements).
- In the meeting in week 4, the focus is on the research plan, which is a crucial element in the overall research process. It should contain the motivation for the central problem statement and a plan how to do the actual research.
- Clearly, all members of the supervision group should receive each other's' draft research plan in time, that is, a few days before the meeting itself; to make the meeting as useful as possible, everyone should have carefully read the different research plans, should have given each of these a thought and should have prepared some questions or suggestions.

Expected output

- In each presentation (meeting 4, meeting 6, meeting 8) session, every student gives a short and focused presentation of his/her research project – corresponding to the stage the project is in – and elaborates both on the contents and the used research method and methodology, on choices that have been made and on solved or unsolved problems they have encountered in their work.
- Each presentation is followed by a general discussion where all participants are supposed to contribute by asking questions for further explanation as well as critical questions about the actual research and by providing constructive suggestions and comments. *This feedback needs to be written down in a document and handed in at the start of each session*
- **Final Paper Due:** 26 June 2025

Contact

- You can always contact me at a.h.machielsen@uu.nl
- Also: a **Microsoft Teams group**
- And I can send you the data in due time (Microsoft Teams files)

Data Sources

- World Values Survey: <https://www.worldvaluessurvey.org> - Geographically coded information about norms & values
- Demographics & Health Survey: <https://dhsprogram.com/data/> - Also geocoded information about D&H
- PRIO GRID Gridded Data
- Eurostat: <https://ec.europa.eu/eurostat/web/main/data/database> - Database about European countries on aggregate (country) and more disaggregated levels
- [Waar Staat Je Gemeente?](#): Dutch Municipality Data
- [Statistiques Locales \(FR\)](#): French Municipality Data
- Global Climate Database: <http://www.worldclim.org/>
- My own website data overview (includes municipality data): [here](#)
- Clio-Infra: <https://github.com/basm92/Clio>
- Standard scholarly measures of politics: <https://github.com/xmarquez/democracyData>
- Correlates of war, armed conflicts: <https://rdr.io/cran/peacesciencer/>
- Political protests: <https://acleddata.com/>
- Labor conflicts: <https://datasets.iisg.amsterdam/dataverse/labourconflicts>
- World Bank, IMF, Countries' National Statistical Agencies, Countries' Parliaments
- [Rspatialdata](#)

References & Literature

Acemoglu, D., Cantoni, D., Johnson, S., & Robinson, J. A. (2011). The consequences of radical reform: The French Revolution. *American economic review*, 101(7), 3286-3307.

Beach, B., & Hanlon, W. W. (2022). Culture and the historical fertility transition. *The Review of Economic Studies*.

Dell, M. (2010). The persistent effects of Peru's mining mita. *Econometrica*, 78(6), 1863-1903.

Dell, M., & Olken, B. A. (2020). The development effects of the extractive colonial economy: The dutch cultivation system in java. *The Review of Economic Studies*, 87(1), 164-203.

Lowes, S., & Montero, E. (2021). Concessions, violence, and indirect rule: evidence from the Congo Free State. *The Quarterly Journal of Economics*, 136(4), 2047-2091.

Lowes, S. (2022). Kinship Structure and the Family: Evidence from the Matrilineal Belt (No. w30509). National Bureau of Economic Research.

References & Literature

Jones, B. F., & Olken, B. A. (2005). Do leaders matter? National leadership and growth since World War II. *The Quarterly Journal of Economics*, 120(3), 835-864.

Voth, H. J. (2021). Persistence—myth and mystery. In *The handbook of historical economics* (pp. 243-267). Academic Press.

Voigtländer, N., & Voth, H. J. (2012). Persecution perpetuated: the medieval origins of anti-Semitic violence in Nazi Germany. *The Quarterly Journal of Economics*, 127(3), 1339-1392.

Besley, T., Burgess, R., Khan, A., & Xu, G. (2022). Bureaucracy and development. *Annual Review of Economics*, 14(1), 397-424.

Beyer, R. C., Franco-Bedoya, S., & Galdo, V. (2021). Examining the economic impact of COVID-19 in India through daily electricity consumption and nighttime light intensity. *World Development*, 140, 105287.

References & Literature

Donaldson, D., & Storeygard, A. (2016). The view from above: Applications of satellite data in economics. *Journal of Economic Perspectives*, 30(4), 171-198.

Castells-Quintana, D., Dienesch, E., & Krause, M. (2021). Air pollution in an urban world: A global view on density, cities and emissions. *Ecological Economics*, 189, 107153.

Felbermayr, G., Gröschl, J., Sanders, M., Schippers, V., & Steinwachs, T. (2022). The economic impact of weather anomalies. *World Development*, 151, 105745.

Beach, B., & Hanlon, W. W. (2018). Coal smoke and mortality in an early industrial economy. *The Economic Journal*, 128(615), 2652-2675.

Hanlon, W. W. (2020). Coal smoke, city growth, and the costs of the industrial revolution. *The Economic Journal*, 130(626), 462-488.